

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

M2 Practical-8 Mini Project

Roll No.: T025	Name: Afsha Amjad Mujawar
Class: TYIT	Batch:1
Date of Assignment: 26-09-2026	Date/Time of Submission: 26-09-2026

Project Title: College Canteen Pre-Order System

Develop a web-based College Canteen Pre-Order System. It should allow students to:

1. View the available food items and their prices.
2. Enter student details such as name, roll number, and course.
3. Select a food item and quantity.
4. Place an order using a POST API.
5. Store the order details in MongoDB.
6. Retrieve the stored orders using a GET API.
7. Display the available food items and placed orders on the React page.

APIs to be implemented

- GET /food – Retrieve food items.
- POST /orders – Add a new food order.
- GET /orders – Retrieve placed orders.

Technologies

- React
- Axios
- Node.js
- Express.js
- MongoDB

Expected Output

The final application should contain:

- Food Item List
- Student Order Form
- Place Order button
- Order List/Table

Design the interface and decide the appropriate fields, MongoDB document structure, and API flow.

Code:

[Server.js:](#)

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
```

```
const app = express();
```

```
app.use(cors());
app.use(express.json());

mongoose.connect("mongodb://127.0.0.1:27017/college")
  .then(() => console.log("MongoDB Connected"));

const Order = mongoose.model("students", new mongoose.Schema({
  studentName: String,
  rollNo: Number,
  course: String,
  foodItem: String,
  price: Number,
  quantity: Number,
  total: Number
}));

const food = [
  { name: "Pizza", price: 100 },
  { name: "Veg Burger", price: 60 },
  { name: "Sandwich", price: 50 },
  { name: "Samosa", price: 20 }
];

// GET /food
app.get("/food", (req, res) => {
  res.json(food);
});

// POST /orders
app.post("/orders", async (req, res) => {
  const { studentName, rollNo, course, foodItem, quantity } = req.body;

  const item = food.find(f => f.name === foodItem);
  const total = item.price * quantity;

  const order = new Order({
    studentName,
    rollNo,
    course,
    foodItem,
    price: item.price,
    quantity,
    total
  });

  await order.save();
  res.json(order);
});

// GET /orders
```

```
app.get("/orders", async (req, res) => {  
  const orders = await Order.find();  
  res.json(orders);  
});  
  
app.listen(5000, () => console.log("Server running on 5000"));
```

App.jsx:

```
import { useEffect, useState } from "react";  
import axios from "axios";  
import "./App.css";
```

```
function App() {  
  const [food, setFood] = useState([]);  
  const [orders, setOrders] = useState([]);  
  
  const [form, setForm] = useState({  
    studentName: "",  
    rollNo: "",  
    course: "",  
    foodItem: "",  
    quantity: 1  
  });  
  
  useEffect(() => {  
    axios.get("http://localhost:5000/food")  
      .then(res => setFood(res.data));  
  
    getOrders();  
  }, []);  
  
  const getOrders = () => {  
    axios.get("http://localhost:5000/orders")  
      .then(res => setOrders(res.data));  
  };  
  
  const change = e => {  
    setForm({ ...form, [e.target.name]: e.target.value });  
  };  
  
  const order = e => {  
    e.preventDefault();  
  
    axios.post("http://localhost:5000/orders", {  
      ...form,
```

```

    rollNo: Number(form.rollNo),
    quantity: Number(form.quantity)
  }).then(() => {
    alert("Order Placed!");
    getOrders();

    setForm({
      studentName: "",
      rollNo: "",
      course: "",
      foodItem: "",
      quantity: 1
    });
  });
};

return (
  <div className="container">

    <h1>College Canteen Pre-Order System</h1>

    <h2>Food Items</h2>

    <div className="food">
      {food.map(f => (
        <div>
          <b>{f.name}</b>
          <p>₹ {f.price}</p>
        </div>
      ))}
    </div>

    <h2>Student Order Form</h2>

    <form onSubmit={order}>
      <input
        name="studentName"
        placeholder="Student Name"
        value={form.studentName}
        onChange={change}
        required
      />

      <input

```

```
    name="rollNo"
    placeholder="Roll No"
    value={form.rollNo}
    onChange={change}
    required
  />
```

```
<input
  name="course"
  placeholder="Course"
  value={form.course}
  onChange={change}
  required
/>
```

```
<select
  name="foodItem"
  value={form.foodItem}
  onChange={change}
  required
>
  <option value="">Select Food</option>
  {food.map(f => (
    <option value={f.name}>
      {f.name} - ₹ {f.price}
    </option>
  ))}
</select>
```

```
<input
  type="number"
  name="quantity"
  min="1"
  value={form.quantity}
  onChange={change}
/>
```

```
<button>Place Order</button>
</form>
```

```
<h2>Placed Orders</h2>
```

```
<table>
  <thead>
```

```

    <tr>
      <th>Name</th>
      <th>Roll No</th>
      <th>Course</th>
      <th>Food</th>
      <th>Price</th>
      <th>Qty</th>
      <th>Total</th>
    </tr>
  </thead>

  <tbody>
    {orders.map(o => (
      <tr>
        <td>{o.studentName}</td>
        <td>{o.rollNo}</td>
        <td>{o.course}</td>
        <td>{o.foodItem}</td>
        <td>₹{o.price}</td>
        <td>{o.quantity}</td>
        <td>₹{o.total}</td>
      </tr>
    ))}
  </tbody>
</table>

</div>
);
}

export default App;

App.css:
body {
  font-family: Arial;
  background: #f2f2f2;
}

.container {
  width: 90%;
  margin: auto;
}

h1 {
```

```
text-align: center;
color: #333;
}
```

```
.food {
display: flex;
gap: 15px;
}
```

```
.food div {
background: white;
padding: 15px;
width: 150px;
border-radius: 8px;
}
```

```
form {
background: white;
padding: 20px;
margin: 20px 0;
}
```

```
input, select, button {
display: block;
width: 100%;
padding: 10px;
margin: 8px 0;
box-sizing: border-box;
}
```

```
button {
background: green;
color: white;
border: none;
cursor: pointer;
}
```

```
table {
width: 100%;
background: white;
border-collapse: collapse;
}
```

```
th, td {
```

```
padding: 10px;
border: 1px solid #ccc;
text-align: center;
}
```

```
th {
background: #333;
color: white;
}
```

Output:

Name	Roll No	Course	Food	Price	Qty	Total
Saqlain	13	B.Sc. IT	Pizza	₹100	2	₹200
Ritesh	5	B.Sc. IT	Veg Burger	₹60	1	₹60
Himani	16	B.Sc. IT	Sandwich	₹50	3	₹150
Reze	14	B.Sc. IT	Samosa	₹20	4	₹80
Makima	16	CS	Cold Coffee	₹70	2	₹140
ritesh	4	IT	Pizza	₹100	1	₹100